

NEUBRANDENBURG, Germany

WMO: 102810

Lat: 53.5997N

Lon: 13.3039E

Elev: 69

StdP: 100.49

Time Zone: 1.00 (EUC)

Period: 95-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-8.8	-14.1	1.1	-10.4	-11.0	1.5	-8.3	13.1	5.1	11.5	5.0	3.0	90	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	29.1	19.0	27.4	18.5	25.4	17.8	20.3	26.6	19.4	25.3	18.5	23.9	3.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.2	22.3	17.2	12.4	21.5	16.4	11.7	20.8	58.6	26.6	55.3	25.3	52.5	23.9	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.9	7.9	DB	-14.7	33.2	4.2	2.2	-17.7	34.7	-20.1	36.0	-22.5	37.2	-25.5	38.8	(4)
(5)			WB	-14.5	22.4	3.9	1.1	-17.4	23.2	-19.7	23.9	-21.9	24.5	-24.7	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	0.2	1.3	4.1	8.9	12.9	16.0	18.6	18.6	14.6	9.6	4.7	1.0	(6)	
	DBStd	7.62	4.86	4.75	3.94	3.99	3.54	3.13	3.03	2.76	3.05	3.63	3.69	4.61	(7)	
	HDD10.0	1306	303	244	187	67	11	0	0	0	2	52	161	278	(8)	
	HDD18.3	3416	561	477	442	285	171	82	34	31	117	271	408	536	(9)	
	CDD10.0	1036	0	1	3	33	102	182	266	267	140	39	3	0	(10)	
	CDD18.3	105	0	0	0	1	4	14	42	39	5	0	0	0	(11)	
	CDH23.3	936	0	0	0	12	57	145	374	301	46	1	0	0	(12)	
	CDH26.7	236	0	0	0	2	10	37	110	72	5	0	0	0	(13)	
(14) Wind	WSAvg	3.8	4.1	4.5	4.3	4.0	3.6	3.6	3.6	3.4	3.5	3.7	3.7	3.9	(14)	
(15) Precipitation	PrecAvg	569	43	34	38	32	52	64	67	62	48	42	43	45	(15)	
	PrecMax	764	80	73	81	67	106	144	228	161	144	78	94	86	(16)	
	PrecMin	398	1	6	10	3	11	19	10	22	16	14	3	9	(17)	
	PrecStd	80	22	14	19	15	24	28	42	33	28	19	19	19	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	12.6	17.2	25.0	27.9	30.3	32.3	31.0	27.1	20.2	14.2	11.7	(19)	
		MCWB	9.1	9.3	11.3	14.8	17.9	19.7	20.6	19.3	17.4	15.4	11.6	9.2	(20)	
	2%	DB	9.5	10.8	14.1	20.8	24.7	27.1	29.5	28.6	24.5	18.1	12.7	9.7	(21)	
		MCWB	7.7	8.4	9.2	13.3	16.7	18.7	19.6	18.6	16.9	14.0	10.4	8.0	(22)	
	5%	DB	7.9	9.1	11.9	18.2	22.1	24.6	27.4	26.8	22.2	16.1	11.1	8.1	(23)	
		MCWB	6.5	7.2	8.6	11.9	15.2	17.4	18.9	18.3	15.8	13.1	9.3	6.7	(24)	
	10%	DB	6.5	7.8	9.9	16.0	20.0	22.5	25.1	24.7	20.0	14.7	9.8	6.7	(25)	
		MCWB	5.2	5.9	7.1	10.7	14.0	16.3	17.9	17.6	15.1	12.3	8.3	5.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	10.1	11.7	16.3	19.3	21.0	21.9	21.4	18.7	16.1	12.3	9.7	(27)	
		MCDB	11.1	11.5	16.0	22.7	25.8	26.7	29.5	27.9	24.4	19.9	13.8	10.8	(28)	
	2%	WB	7.8	8.7	10.2	14.0	17.5	19.4	20.5	20.0	17.6	14.4	10.7	8.1	(29)	
		MCDB	9.3	10.5	13.2	20.2	23.4	25.6	27.0	25.7	22.6	17.1	12.2	9.4	(30)	
	5%	WB	6.5	7.4	8.6	12.4	16.0	18.0	19.5	19.0	16.6	13.5	9.4	6.7	(31)	
		MCDB	7.7	9.1	11.5	17.2	21.2	23.3	25.7	24.2	20.8	15.9	10.8	7.9	(32)	
	10%	WB	5.3	5.9	7.2	11.0	14.6	16.8	18.6	18.2	15.6	12.4	8.4	5.5	(33)	
		MCDB	6.4	7.6	9.6	15.3	19.1	21.6	24.1	23.2	19.5	14.6	9.6	6.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.4	7.1	9.6	10.4	9.8	9.5	9.6	8.5	6.7	4.7	4.1	(35)	
		MCDBR	5.2	5.8	10.2	13.2	14.1	13.5	13.7	13.3	11.8	8.4	5.5	4.5	(36)	
	5% WB	MCWBR	4.4	4.3	6.5	7.3	7.5	6.8	5.7	5.7	5.8	5.2	4.2	4.0	(37)	
		MCWBR	4.9	5.5	9.5	12.2	12.8	12.3	12.2	11.2	10.6	8.0	5.3	4.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.313	0.338	0.371	0.403	0.400	0.397	0.417	0.413	0.378	0.360	0.336	0.307	(40)	
		taud	2.316	2.309	2.254	2.229	2.279	2.310	2.275	2.297	2.372	2.387	2.367	2.314	(41)	
	Ebn at Noon		647	751	801	823	848	853	826	806	790	717	615	577	(42)	
		Edh at Noon	59	82	106	124	124	122	124	115	94	75	57	50	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.59	1.25	2.44	4.08	5.11	5.39	5.02	4.22	3.08	1.65	0.71	0.41	(44)	
	RadStd		0.07	0.18	0.27	0.57	0.61	0.42	0.69	0.34	0.33	0.22	0.08	0.04	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+1.03	N/A	N/A	N/A	N/A	(46)
(47)	Regional (41 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page